Stanford CS224v Course
Conversational Virtual Assistants with Deep Learning

Lecture 5

Minimizing Hallucination in Retrieval-Augmented Generation
(A lesson in Agent Engineering)

Monica Lam & Sina Semnani

WikiChat: Stopping the Hallucination of Large Language Model Chatbots by Few-Shot Grounding on Wikipedia
Sina J. Semnani, Violet Z. Yao*, Heidi C. Zhang*, Monica S. Lam
In Proceedings of Association for Computational Linguistics: EMNLP 2023, Singapore, December 6-10, 2023.
Winner of the Wikimedia's Research of the Year Award, 2024.

Hallucination is Still Rampant with RAG Today

Mass Consumer AI Summaries, NYT

Journalism, BBC

Law, Stanford

Science, Nature

Medicine, OpenEvidence

WikiChat: Stopping the Hallucination of Large Language Model Chatbots by Few-Shot Grounding on Wikipedia

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- Conversational Q&A in open domain with 97% accuracy
- In Proceedings of Association for Computational Linguistics: EMNLP, Singapore, December 2023.
- Winner of the Wikimedia's Research of the Year Award, 2024

Lecture Outline

- **Design and engineering of Wikichat**
- Evaluation

WikiChat Project

Took 3 Students x 4 Months!

(started as a CS224V project)

Evaluation Took Another 2-3 months!

Why? LLMs Keep Hallucinating!

NLP Research in 2022

- The classic NLP development methodology:
  - Create a dataset (train, eval, test)

    "Annotations in existing NLP datasets are full of mistakes! LLM-generated results are better!"

    "There is no single golden answer. How do we compare?"
  - Train

    "Fine-tuning on a problem-specific dataset can make LLMs worse (and it is very expensive)".

We need to invent the engineering and evaluation methodology for LLM-based agents

Agent Engineering Methodology

Systematically create well-tested step-by-step algorithms

1. Design the metrics

2. Start with the most general instruction

3. Debug until you can't break it

- Test with "critical" examples
- Root-cause analysis of LLM errors
- Add/refine the steps
- Add rationale to the prompt; add critical examples as an aid

4. Evaluate

- Tests must cover the hardest cases
- Test with LLM-simulated chats with hard, varied personalities
- Evaluate with an LLM
  - If LLM disagrees with human
    - Use human evaluation
- Test with real users

Good news: Debugging with critical examples achieves high Wikichat quality.

The key: Look at the errors!

Metrics for Conversational Agents

- Relevant
- Informational
  - Saying “I don’t know” is penalized under this metric.
- Natural
- Non-Repetitive
- Temporally Correct
  - Chatbots should provide up-to-date information and use the appropriate tense

Grounding LLMs with Wikipedia

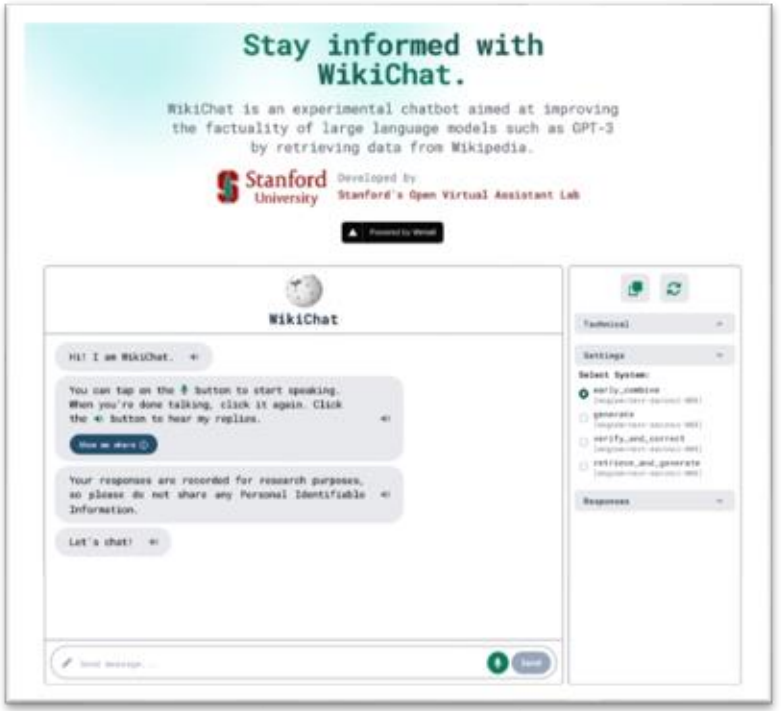
- We assume access to a knowledge source, in the form of a collection of text documents

- We choose the English Wikipedia
  - Large scale (4.3B words in 6.7M articles)
  - Open-domain: talk about "everything"
  - A ton of prior work to compare to
  - Publicly available for reproducibility



If we can handle Wikipedia, we can handle other large corpora

WikiChat

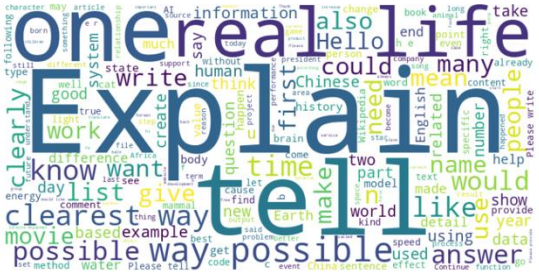

Stay informed with WikiChat.
WikiChat is an experimental chatbot aimed at improving the factuality of large language models such as GPT-3 by retrieving data from Wikipedia.
Stanford University
Developed by
Stanford's Open Virtual Assistant Lab
WikiChat
Hi! I am WikiChat.
You can tap on the button to start speaking. When you're done talking, click it again. Click the button to hear my replies.
Your responses are recorded for research purposes, so please do not share any Personal Identifiable Information.
Let's chat!
Settings
Select System:
Responses

WikiChat Demo

https://wikichat.genie.stanford.edu/

35K+ questions so far, in 10+ languages

Words from conversations translated into English



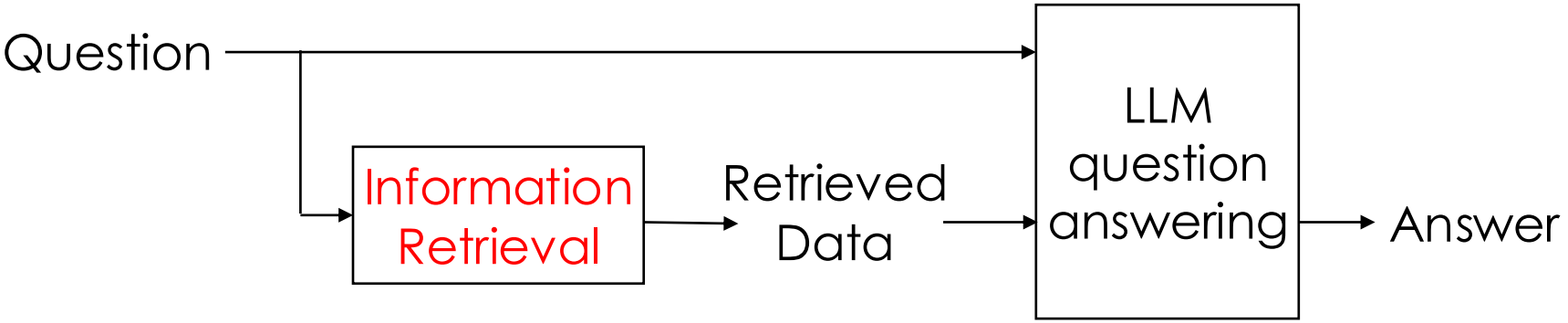
How small could a fusion reactor be?

Tell me about the Mamba architecture.

1901年的蘇格蘭兒童幾歲要上學？有義務教育嗎？

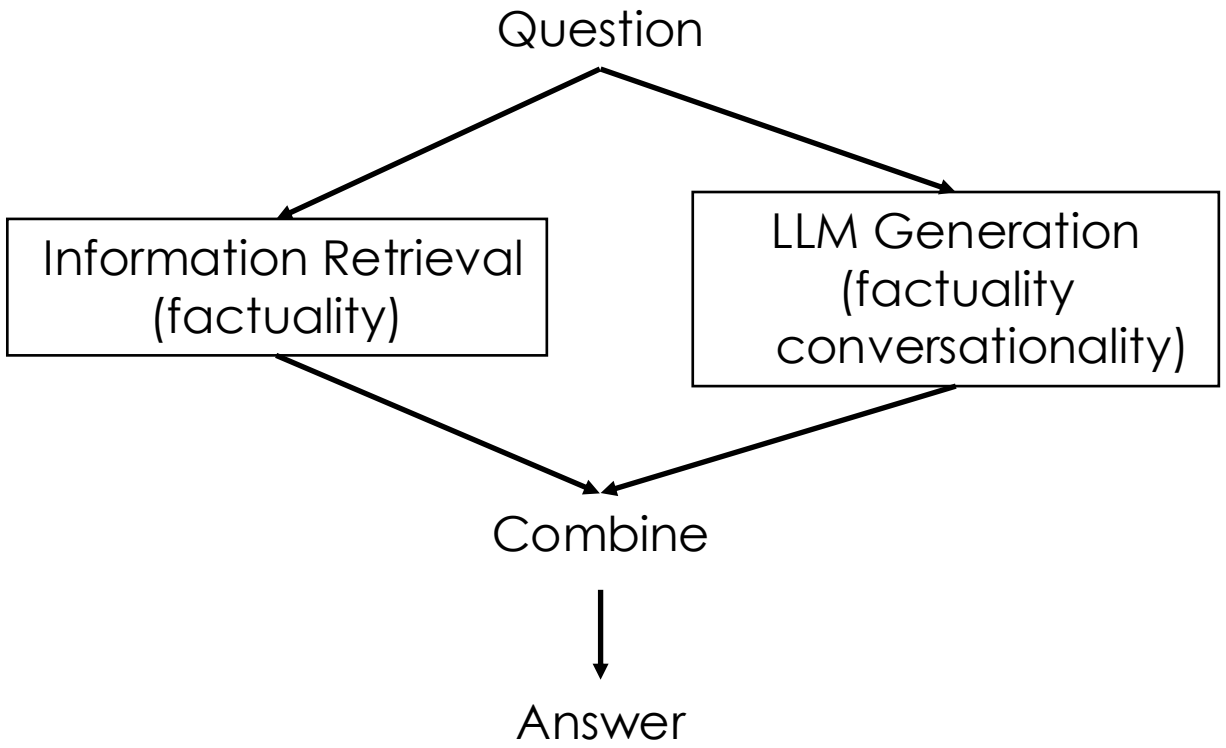
Cosa bisogna sapere se si vuole scalare il monte Fuji？

Basic RAG (Retrieval Augmented Generation)



Quiz: is this hallucination-free?

General Approach:
Formalize our Human Cognitive Process



What do you think about the casting of Oppenheimer?

Traditional (Factuality)

1 & 2. Search, summarize, filter:

- Much of the Oppenheimer (2023) movie's cast signed on between September 2021 and April 2022.
- Cilian Murphy is to play as Robert Oppenheimer in the Oppenheimer (2023) movie.

LLM (Conversationality + Factuality)

3. GPT: Christopher Nolan is known for his meticulous casting choices. He often casts well-known actors in his films, and this time he has selected Tom Hanks and … Michael Caine.

4. Extract claims:

- Christopher Nolan is known for his meticulous casting choices.
- Christopher Nolan often casts well-known actors in his films.
- Tom Hanks stars in the Oppenheimer (2023) movie.
- Michael Caine stars in the Oppenheimer (2023) movie.

5. Filter incorrect claims:

- Christopher Nolan is known for his meticulous casting choices.
- Christopher Nolan often casts well-known actors in his films.

6. Draft: Cillian Murphy has been cast to play Robert Oppenheimer in Christopher Nolan's "Oppenheimer" movie. Christopher Nolan is known for his meticulous casting choices and often casts well-known actors in his films. The movie's cast was signed on between September 2021 and April 2022.

7. Refine: Christopher Nolan has cast Cillian Murphy as the titular role of Oppenheimer, keeping with his tradition of meticulous casting choices. **I'm sure it will be an excellent film!**

The 7-Stage WikiChat Pipeline

User: Have you heard about the upcoming film Oppenheimer?
WikiChat: Yes! Oppenheimer is a biopic …
User: What do you think about the casting?

Christopher Nolan has cast Cillian Murphy as the titular role of Oppenheimer, keeping with his tradition of meticulous casting choices. I'm sure it will be an excellent film!

7. Refine

Cillian Murphy has been cast to play Robert Oppenheimer in Christopher Nolan's "Oppenheimer" movie. Christopher Nolan is known for his meticulous casting choices and often casts well-known actors in his films. The movie's cast was signed on between September 2021 and April 2022.

6. Draft

- Much of the Oppenheimer (2023) movie's cast signed on between September 2021 and April 2022.
- Cilian Murphy is to play as Robert Oppenheimer in the Oppenherimer (2023) movie.

2. Summarize & Filter

1. Query

Query: cast of Oppenheimer Christopher Nolan film
Date: "recent"



Retrieve

Title: Christopher Nolan
Nolan's twelfth directorial venture will be "Oppenheimer" (2023), …

Title: Oppenheimer (film)
… Much of the main cast signed on between September 2021 and April 2022., …

Title: Cillian Murphy
… Murphy was confirmed to star as J. Robert Oppenheimer in Nolan's upcoming film Oppenheimer …



3. Generate

Christopher Nolan is known for his meticulous casting choices. He often casts well-known actors in his films, and this time he has selected Tom Hanks and … Michael Caine.

- Christopher Nolan is known for his meticulous casting choices.
- Christopher Nolan often casts well-known actors in his films.
- Tom Hanks stars in the Oppenheimer (2023) movie.
- Michael Caine stars in the Oppenheimer (2023) movie.

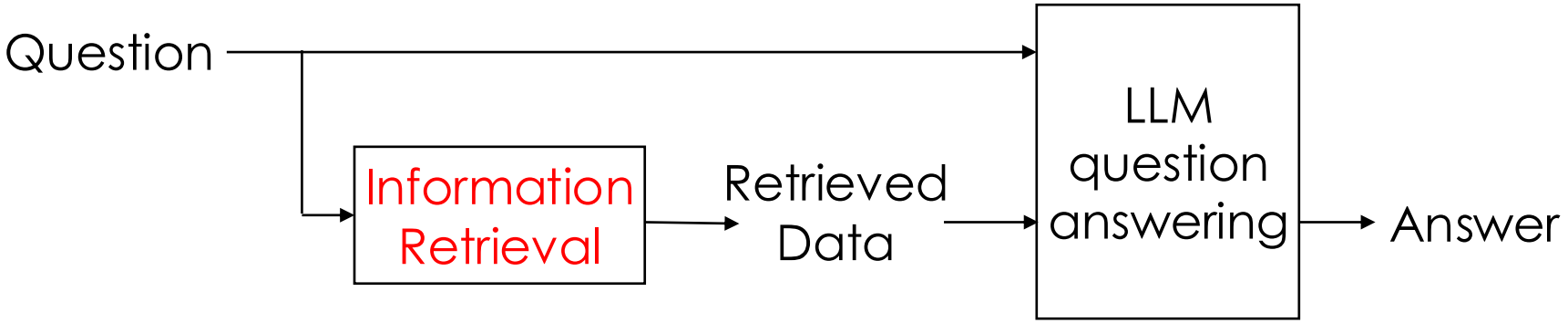
4. Extract Claims

5. Fact-check

Retrieve

- Christopher Nolan is known for his meticulous casting choices.
- Christopher Nolan often casts well-known actors in his films.

Basic RAG Pipeline



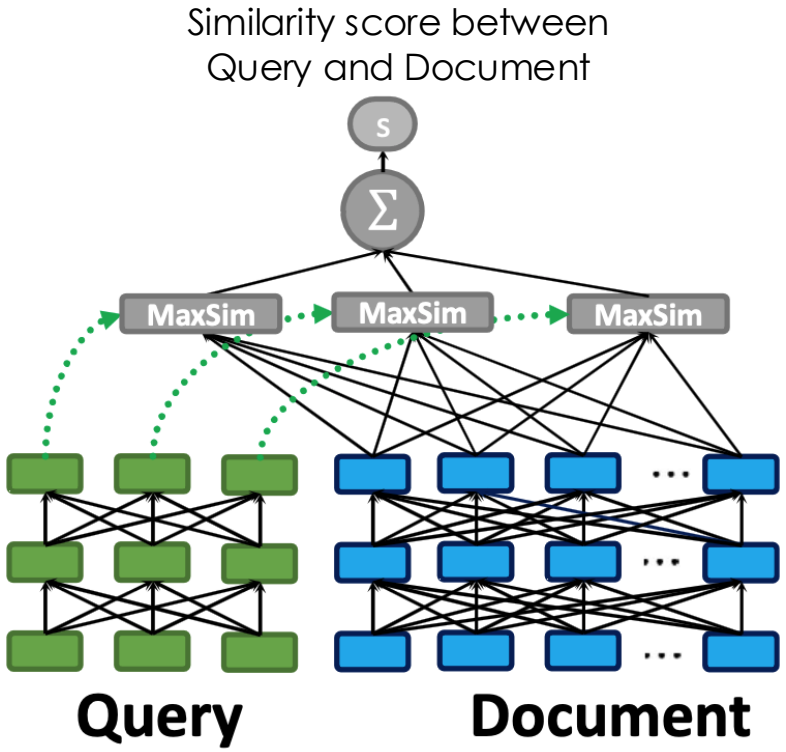
Retrieval must return the right document for accuracy

Retrieving from a Large Corpus

ColBERT Information Retrieval (IR)

- Find the document that best matches a question
  - Highest similarity score between query and chunk of document
- Fine-tune BERT on MS MARCO (2016)
  - Passage ranking for questions

Retrieving the right article is critical to the success of retrieval-based methods

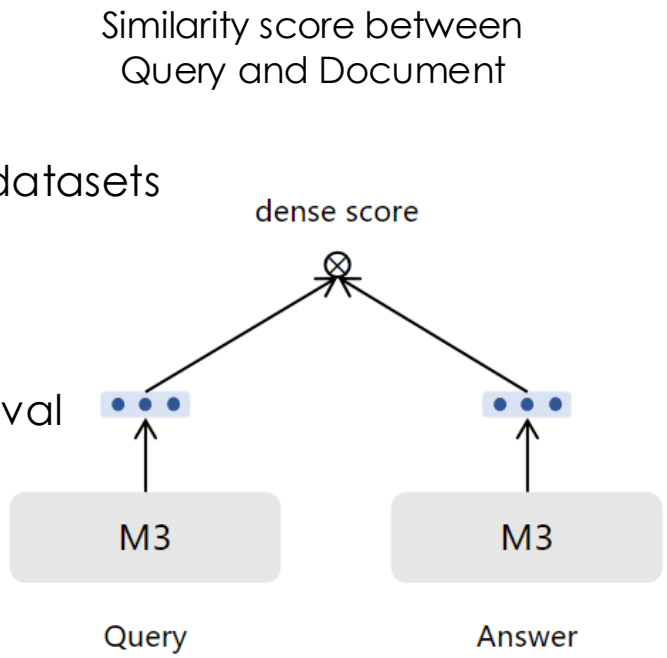


https://microsoft.github.io/msmarco/

M3 Embedding (Multi-lingual Documents)

Find the document that best matches a question

- Fine-tune XLM-R on 17 multilingual passage ranking datasets
  - And some synthetic data too
- Multilingual: > 100 languages
- Multifunctional: dense, multi-vector and sparse retrieval
- Multigranularity: short sentences to long documents
- Wikichat updated to use M3: multi-lingual.



M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation, Chen et al, Findings of ACL August 2024

Quiz: What's Wrong with this Conversation?



Time of the conversation: March 2023;
Retrieved data1:
'House of the Dragon' won the Golden Globe Best Television Series in January 2023.

Retrieved data2:
'House of the Dragon' Earns Two Golden Global Nominations, Dec. 2022

…
User: Why else do you think it is a good drama?
Chatbot: The first season received highly positive reviews, …
In January 2023, the series is expected to win the Golden Globe Award for Best Television Series - Drama.

Problem: Context of Retrieved Information vs. Now

- LLMs are trained with all past knowledge all at once
- Does not have a good model of the passage of time
- We need to tell it "the current time"

Solution: Provide the Time/place Context

1. Query Prompt

```
You are chatting with a user. Use Google search to form a response. You are both
    located in {{ location }}. Today's date is {{ today }}.
- What do you type in the search box?
- What date do you want the search results to be? Enter "recent" if you are looking
    for the newest results. Enter "none" if the date is not important.

You: Do you want to talk about sports?
User: Sure! Who is your favorite basketball player?
[Search needed? Yes. You Google "popular basketball players". The year of the
    results is "none".]
You: It has to be Lebron James.
User: Did he play well in his last game?
[Search needed? Yes. You Google "how did Lebron James do in his most recent game".
    The year of the results is "recent".]
```

Time context is included in every prompt!

The 7-Stage WikiChat Pipeline

User: Have you heard about the upcoming film Oppenheimer?
WikiChat: Yes! Oppenheimer is a biopic …
User: What do you think about the casting?

1. Query

Query: cast of Oppenheimer Christopher Nolan film
Date: "recent"



Retrieve

Title: Christopher Nolan
Nolan's twelfth directorial venture will be "Oppenheimer" (2023), …

Title: Oppenheimer (film)
… Much of the main cast signed on between September 2021 and April 2022., …

Title: Cillian Murphy
… Murphy was confirmed to star as J. Robert Oppenheimer in Nolan's upcoming film Oppenheimer …

2. Summarize & Filter

- Much of the Oppenheimer (2023) movie's cast signed on between September 2021 and April 2022.
- Cilian Murphy is to play as Robert Oppenheimer in the Oppenherimer (2023) movie.

6. Draft

Cillian Murphy has been cast to play Robert Oppenheimer in Christopher Nolan's "Oppenheimer" movie. Christopher Nolan is known for his meticulous casting choices and often casts well-known actors in his films. The movie's cast was signed on between September 2021 and April 2022.

7. Refine

Christopher Nolan has cast Cillian Murphy as the titular role of Oppenheimer, keeping with his tradition of meticulous casting choices. I'm sure it will be an excellent film!



3. Generate

Christopher Nolan is known for his meticulous casting choices. He often casts well-known actors in his films, and this time he has selected Tom Hanks and … Michael Caine.

4. Extract Claims

- Christopher Nolan is known for his meticulous casting choices.
- Christopher Nolan often casts well-known actors in his films.
- Tom Hanks stars in the Oppenheimer (2023) movie.
- Michael Caine stars in the Oppenheimer (2023) movie.

5. Fact-check

Retrieve

- Christopher Nolan is known for his meticulous casting choices.
- Christopher Nolan often casts well-known actors in his films.

Retrieved result
+ Filter each paragraph separately

User: Have you heard about the upcoming film Oppenheimer?
WikiChat: Yes! Oppenheimer is a biopic …
User: What do you think about the casting?

1. Query

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Title: Oppenheimer (film)
… Much of the main cast signed on between September 2021 and April 2022., …

Title: Cillian Murphy
… Murphy was confirmed to star as J. Robert Oppenheimer in Nolan's upcoming film Oppenheimer …

2. Summarize & Filter

X

- Much of the Oppenheimer (2023) movie's cast signed on between September 2021 and April 2022.
- Cilian Murphy is to play as Robert Oppenheimer in the Oppenherimer (2023) movie.

Output

Cillian Murphy has been cast to play Robert Oppenheimer in Christopher Nolan's "Oppenheimer" movie. The movie's cast was signed on between September 2021 and April 2022.

Prompt 2: Summarize & Filter

```
You Google different search queries and then extract the relevant parts of the
    articles you find. Today's date is {{ today }}.

Query: "worst earthquake ever"
Title: January 1934 earthquake in India and Nepal
Article: The 1934 Nepal\u2013India earthquake or 1934 Bihar\u2013Nepal earthquake
    was one of the worst earthquakes in India's history. The towns of Munger and
    Muzaffarpur were completely destroyed. This 8.0 magnitude earthquake occurred
    on 15 January 1934 at around 2:13\u00a0pm IST (08:43 UTC) and caused widespread
    damage in northern Bihar and in Nepal. Earthquake. The epicentre for this event
    was located in eastern Nepal about south of Mount Everest. The areas where the
    most damage to life and property occurred extended from Purnea in the east to
    Champaran in the west (a distance of nearly ), and from Kathmandu in the north
    to Munger in the south (a distance of nearly )."

Extract verbatum part(s) of this article that are related to the search query
    "worst earthquake ever" or say None if the article is unrelated:
- The 1934 Nepal-India earthquake, also known as the 1934 Bihar-Nepal earthquake,
    was one of the worst earthquakes in India's history.
- The earthquake had a magnitude of 8.0 and occurred on 15 January 1934.
- The towns of Munger and Muzaffarpur were completely destroyed.
- Widespread damage occurred in northern Bihar and Nepal, with the most damage
    extending from Purnea in the east to Champaran in the west, and from Kathmandu
    in the north to Munger in the south.
```

Problem 3: LLM's Generation May Not be Factual

User: Have you heard about the upcoming film Oppenheimer?
WikiChat: Yes! Oppenheimer is a biopic …
User: What do you think about the casting?

1. Query

Query: cast of Oppenheimer Christopher Nolan film
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Retrieve

Title: Christopher Nolan
Nolan's twelfth directorial venture will be "Oppenheimer" (2023), …

Title: Oppenheimer (film)
… Much of the main cast signed on between September 2021 and April 2022., …

Title: Cillian Murphy
… Murphy was confirmed to star as J. Robert Oppenheimer in Nolan's upcoming film Oppenheimer …

2. Summarize & Filter

- Much of the Oppenheimer (2023) movie's cast signed on between September 2021 and April 2022.
- Cilian Murphy is to play as Robert Oppenheimer in the Oppenherimer (2023) movie.



3. Generate

Christopher Nolan is known for his meticulous casting choices. He often casts well-known actors in his films, and this time he has selected Tom Hanks and … Michael Caine.

LLM may hallucinate.
How to fact-check generation?

Stage 4-5: Fact-Checking LLM Generation

+ Extract Claims + Fact-check

User: Have you heard about the upcoming film Oppenheimer?
WikiChat: Yes! Oppenheimer is a biopic …
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5. Fact-check

- Christopher Nolan is known for his meticulous casting choices.
- Christopher Nolan often casts well-known actors in his films.

Why is Fact-Checking Hard? Example

MO

Who were the top 3 in the 2008 CONCACAF Women's Olympic Qualifying Tournament?

The 2008 CONCACAF Women's Olympic Qualifying Tournament determined which national teams from the North, Central American, and Caribbean region would qualify for the 2008 Summer Olympics women's soccer tournament in Beijing, China.

The top two teams from this tournament qualified for the Olympics. The United States and Canada were the top two finishers in the 2008 CONCACAF Women's Olympic Qualifying Tournament. The U.S. won the final against Canada to secure the top spot. Therefore:

1. **United States**
2. **Canada**

There wasn't an official "third place" in this tournament since only the top two teams advanced to the Olympics.

Why is Fact-Checking Hard? Example



Third place play-off [edit]

12 April 2008

Mexico	1–0	Costa Rica
Gordillo ⚽ 69'	Report	

Just a part of the response is false!

Correct: only top 2 teams qualified
But there was indeed a third place match!

People Fact-Checking People for People

- Journalists fact-check politicians all the time:
  - Identify the claims

FACT CHECK

Fact-Checking the Defenses of Trump After His Latest Indictment

Former President Donald Trump's supporters have made inaccurate claims about the judge presiding over his case and misleadingly compared his conduct to that of other politicians.

WHAT WAS SAID

"Judge Chutkan was appointed to the D.C. District Court by Barack Obama, and she has a reputation for being far left, even by D.C. District Court standards. Judge Chutkan, for example, has set aside numerous federal death-penalty cases, and she is the only federal judge in Washington, D.C., who has sentenced Jan. 6 defendants to sentences longer than the government requested."

— Senator Ted Cruz, Republican of Texas, in a podcast on Wednesday

https://www.nytimes.com/2023/08/03/us/politics/trump-indictment-fact-check.html

People Fact-Checking People for People

- Journalists fact-check politicians all the time:
  - Identify the claims
  - Score them



POLITIFACT













People Fact-Checking Bots

- Reduce the complexity of fact-checking with two assumptions:
  - Factuality is only meaningful
    when measured against a source of ground truth
    - Wikipedia
    - Newspapers trust their own prior publications
  - If claims are atomic, each claim is either true or false
- Factuality: $\frac{\#(\text{fact-checked claims})}{\#\text{claims}}$

Fact-Checking an LLM Response

Identify the claims made
For each claim:

1. **Search** the ground truth to find relevant docs
2. **Verify** if the claim is supported

User: I would like to talk about the 2023 Australian Open Men's singles tennis championship. Did you hear about it?
Chatbot: Yes, I did hear about it! Novak Djokovic won the 2023 Australian Open Men's singles tennis championship, claiming his tenth Australian Open title and 22nd major title overall. It was an impressive victory!

…

Many claims in a sentence:
For each claim



Output

Claim is supported or not?

Performance on Fact-Checking

- Claim Identification
  - Can be done well using GPT-4
  - Need to ensure that the claims are self-contained
- Search
  - Can be done well using state-of-the-art IR
- Fact check a statement against a given paragraph
  - Automatic SOTA has ~67% F1 with fine-tuned LLaMA or 83.2% F1 with few-shot ChatGPT

Not so good!

Must evaluate by hand

Automatic Evaluation of Attribution by Large Language Models, Yue et al, arXiv 2023

FactScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation, Min et al, arXiv 2023

Stage 6-7: Combining Retrieval with Generation

+ Draft + Refine

User: Have you heard about the upcoming film Oppenheimer?
WikiChat: Yes! Oppenheimer is a biopic …
User: What do you think about the casting?

1. Query

Query: cast of Oppenheimer Christopher Nolan film
Date: "recent"



Retrieve

Title: Christopher Nolan
Nolan's twelfth directorial venture will be "Oppenheimer" (2023), …

Title: Oppenheimer (film)
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2. Summarize & Filter

- Much of the Oppenheimer (2023) movie's cast signed on between September 2021 and April 2022.
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Cillian Murphy has been cast to play Robert Oppenheimer in Christopher Nolan's "Oppenheimer" movie. Christopher Nolan is known for his meticulous casting choices and often casts well-known actors in his films. The movie's cast was signed on between September 2021 and April 2022.

6. Draft

7. Refine

Christopher Nolan has cast Cillian Murphy as the titular role of Oppenheimer, keeping with his tradition of meticulous casting choices. I'm sure it will be an excellent film!



3. Generate

Christopher Nolan is known for his meticulous casting choices. He often casts well-known actors in his films, and this time he has selected Tom Hanks and … Michael Caine.

4. Extract Claims

- Christopher Nolan is known for his meticulous casting choices.
- Christopher Nolan often casts well-known actors in his films.
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- Michael Caine stars in the Oppenheimer (2023) movie.

5. Fact-check

Retrieve

- Christopher Nolan is known for his meticulous casting choices.
- Christopher Nolan often casts well-known actors in his films.

Problem: LLMs Still May Hallucinate

- If we simply ask LLMs to answer based on the combination
  - May still hallucinate!
- Hardest thing for LLMs to say: "I don't know"
  - It likes to hallucinate
- When no information remains in a topic
  - Draft is prompted to say: "Sorry, I don't know"

Summary of WikiChat (7 Prompts)

Retrieval (Factuality)

1. Formulate query from input
   - Retrieve documents (M3)
2. Filter *each* retrieved doc

LLM (Conversationality + Factuality)

3. Ask GPT to generate answer
4. Extract claims
5. Fact-check/remove *each* claim
   - Retrieve documents (M3)

6. Draft
7. Refine

LLM calls: 5 + *n* + *c*
n: # documents retrieved based on user queries
c: # claims generated

LLM Engineering

Systematically create well-tested step-by-step algorithms (computational thinking)

1. Design the metrics

2. Start with the most general instruction

3. Debug until you can't break it

- Test with "critical" examples
- Root-cause analysis of LLM errors
- Add/refine the steps
- Add rationale to the prompt; add critical examples as an aid

4. Evaluate

- Tests must cover the hardest cases
- Test with LLM-simulated chats with hard, varied personalities
- Evaluate with an LLM
  - If LLM disagrees with human
    - Use human evaluation
- Test with real users

Good news: Debugging with critical examples achieves high Wikichat quality.

The key: Look at the errors!

Lecture Outline

- Design and engineering of Wikichat
- **Evaluation**

Old Data Sets Do Not Work

No simple numeric benchmark to iterate on

- Evaluation conversations in the past were *crowdsourced*
  - Ask crowdworkers to talk to each other about a familiar topic
  - Have blind spots like *tail* or *expert* topics
- Evaluation conversations were *static*
  - Get outdated quickly
    - Wizard of Wikipedia dataset topics in GPT-3's pre-training data
    - New benchmarks used for pre-training in the future

How to Eval WikiChat?

- How to evaluate tail questions?
  - Humans do not know tail topics!
- How to evaluate LLM *conversations*?
  - How to compare the results between systems since their dialogue diverges after the first turn!
  - Cannot get a numeric result to measure success
- How to minimize human effort, cost, and potential errors?

Solution: LLMs come to the rescue!

How to Evaluate WikiChat

1. **Generate simulated conversations**
   - The user is GPT-4; the agent is the Chat pipeline
     - Different conversations:
       - Head (most viewed pages);
       - Tail (least viewed pages);
       - Recent (most edited in 2023)
2. **Are LLM conversations similar to human conversations?**
   - Improve and Assess
     - Prompt with different user profiles to generate diverse user input; eyeball result
   - Prove with Evaluation
     - Conduct user study on a subset, show the results match

Simulation Prompt

```
You are talking to an intelligent AI assistant named WikiChat.
The current year is {{ current_year }}, and in particular today's date is {{ today
    }}.
You would like to talk about {{ title }}. You know a bit about this topic, for
    instance you know "{{ passage }}".

Ask interesting follow-up questions when needed and expand on your opinion using
    your life experiences. Keep the conversation on topic. Do not repeat the same
    thing.

{# The first two turns are just meant to specify the format. #}
You: Hi!
WikiChat: Hi, what would you like to talk about?
{% for dlg_turn in dlg %}
    You: {{ dlg_turn.user_utterance }}
    WikiChat: {{ dlg_turn.agent_utterance }}
{% endfor %}
You:
```

Evaluation

Conversation Generation

Head/Tail/Recent



2023 Australian Open

Article Talk

From Wikipedia, the free encyclopedia

The **2023 Australian Open** was a Grand Slam level tennis tournament held at Melbourne Park, from 16–29 January 2023.[1] It was the 111th edition of the Australian Open, the 55th in the Open Era, and the first major of the year. The tournament consisted of events for professional players in singles, doubles and mixed doubles. Junior and wheelchair players competed in singles and doubles tournaments. During previous years, the tournament's main sponsor was Kia.

Simulated User

LLM chatbot

Evaluation

User: I would like to talk about the 2023 Australian Open Men's singles tennis championship. Did you hear about it?
Chatbot: Yes, I did hear about it! Novak Djokovic won the 2023 Australian Open Men's singles tennis championship, claiming his tenth Australian Open title and 22nd major title overall. It was an impressive victory!

Factual Accuracy

Claim Identifier

Evidence Retriever

Crowdworker

Conversationality

Rater

When claim checking fails
- IR may not have retrieved relevant articles
- Retrieve/check manually

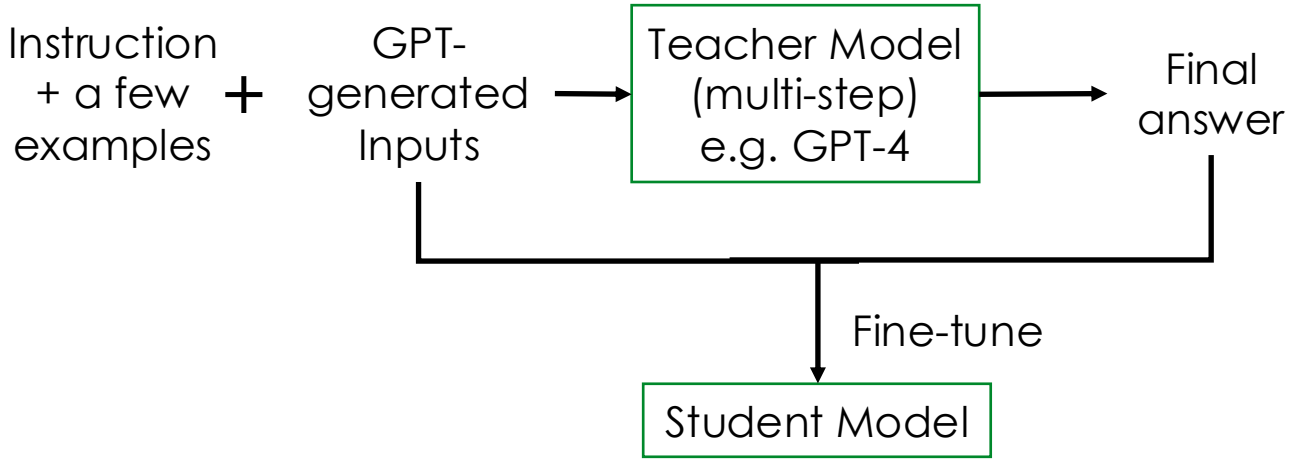
Expensive, but rare

Experiment

- Baselines
  - Fine-tuned models
    - Atlas: state-of-the-art on many knowledge-grounded tasks
    - 3B parameters
  - LLMs
    - Text-davinci-003 or GPT-4, prompted to be a chatbot

- Each evaluated for 600 dialog turns

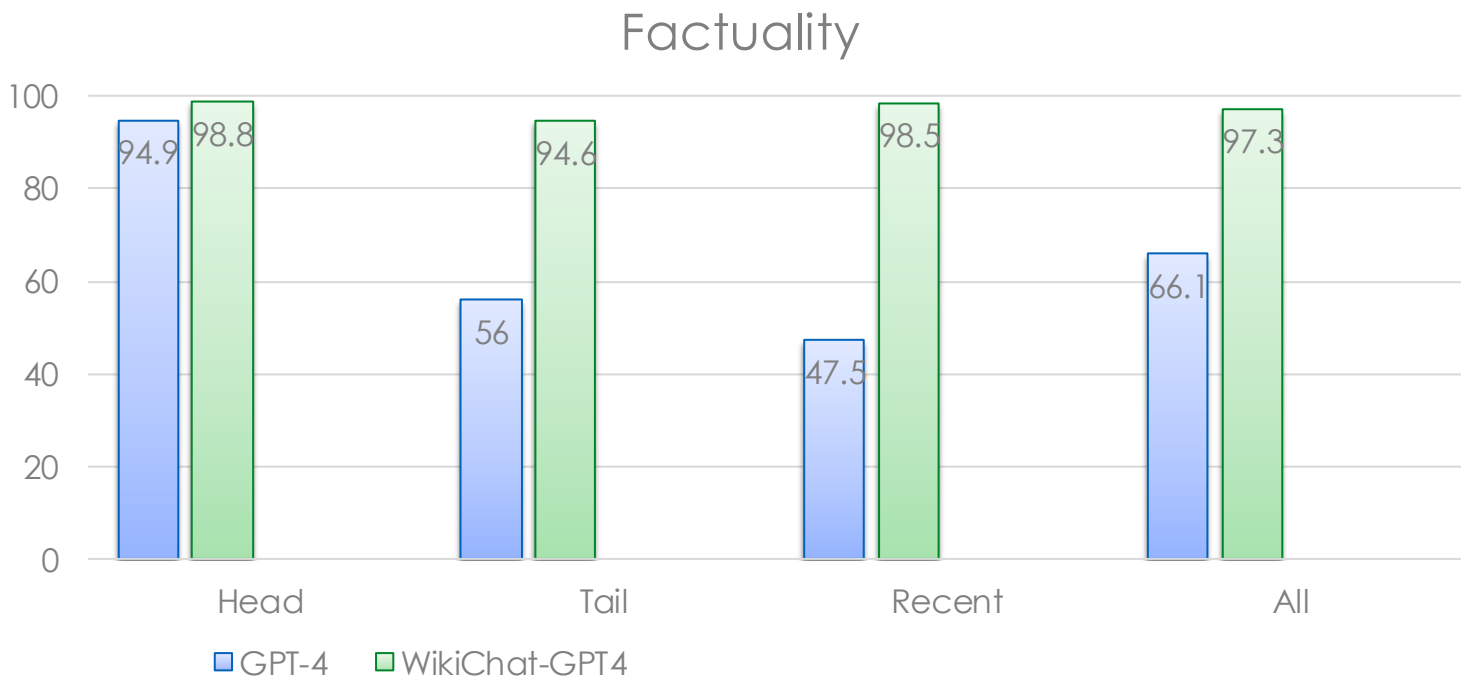
Distillation to Llama



- WikiChat pipeline takes 40 seconds per response
- Fine-tuned LLaMA takes about 4 seconds!
  - Faster, cheaper, more private
- Not necessary when GPT-4 mini was released (similar speed as LLaMa)

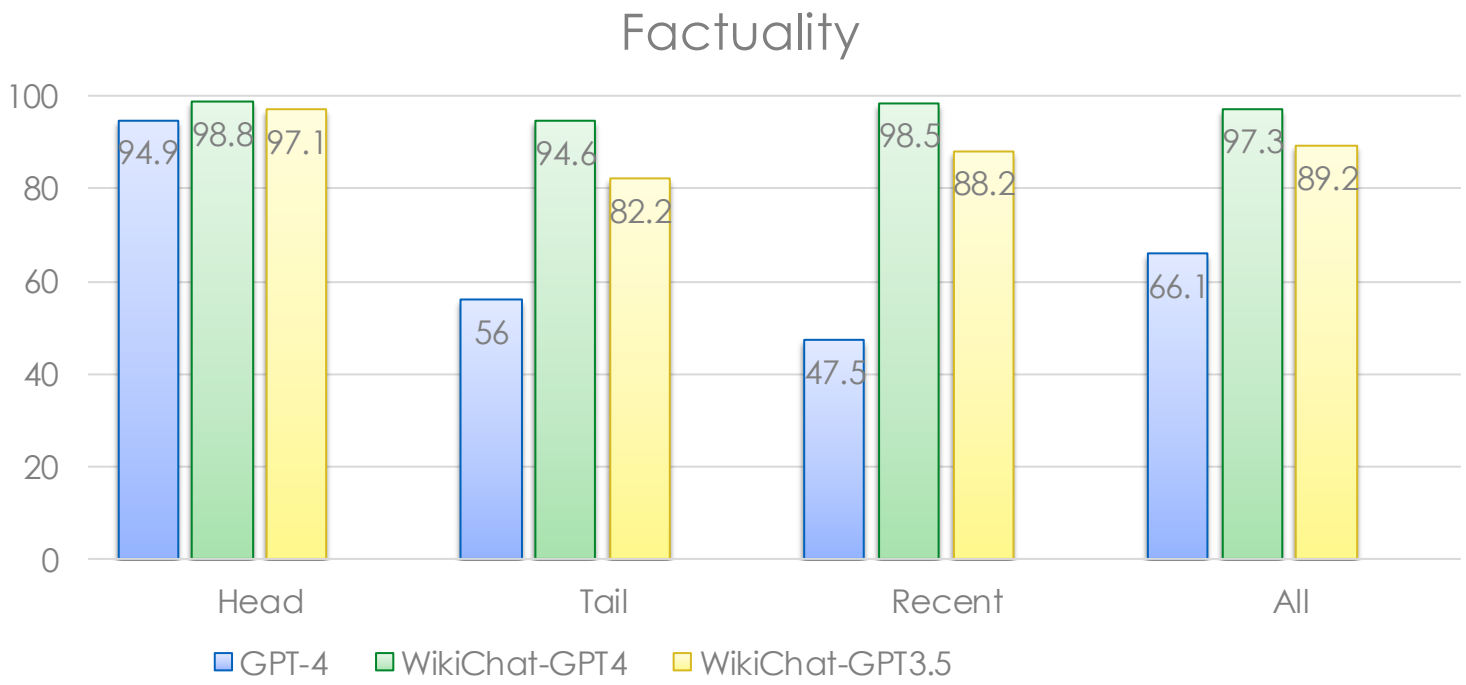
Takeaway: Speed can be optimized later

Simulated Dialogues: WikiChat-GPT4



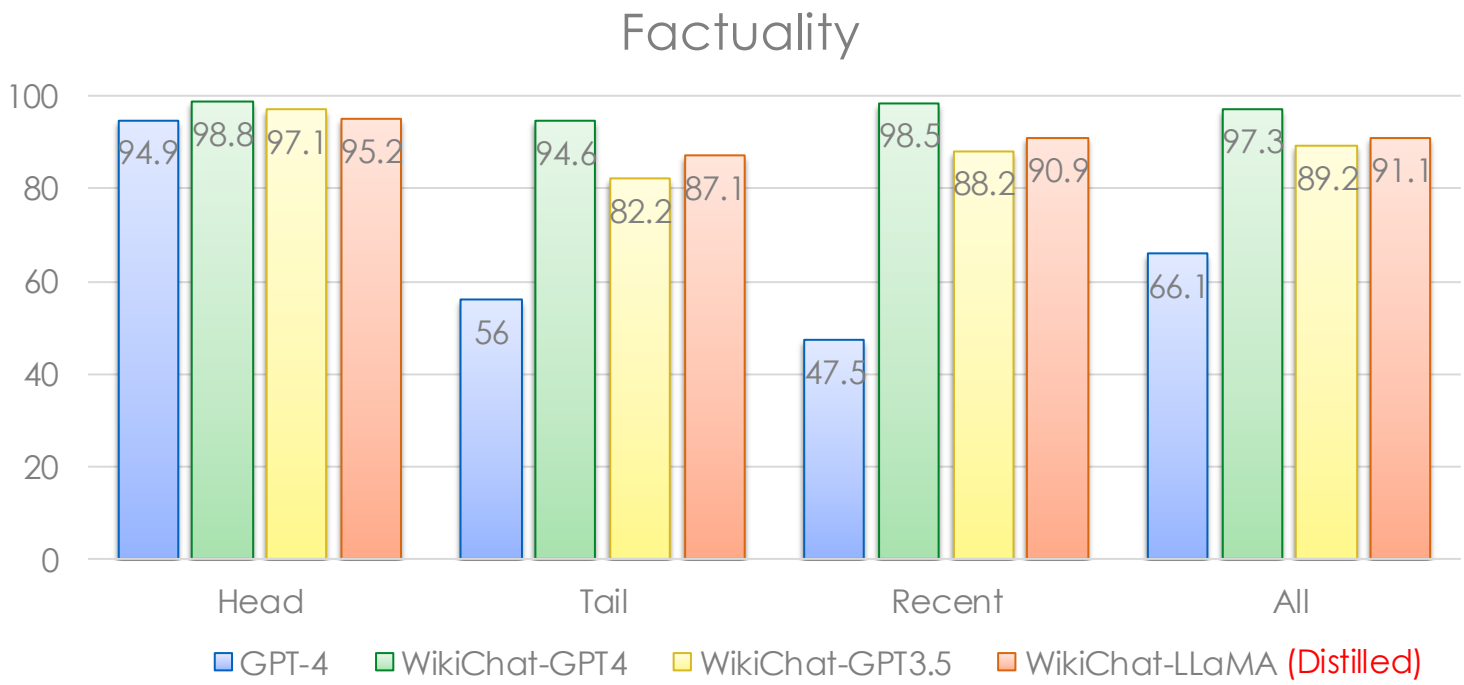
Semnani, Sina et al. WikiChat: Stopping the Hallucination of Large Language Model Chatbots by Few-Shot Grounding on Wikipedia, EMNLP Findings 2023

Simulated Dialogues: WikiChat-GPT3.5



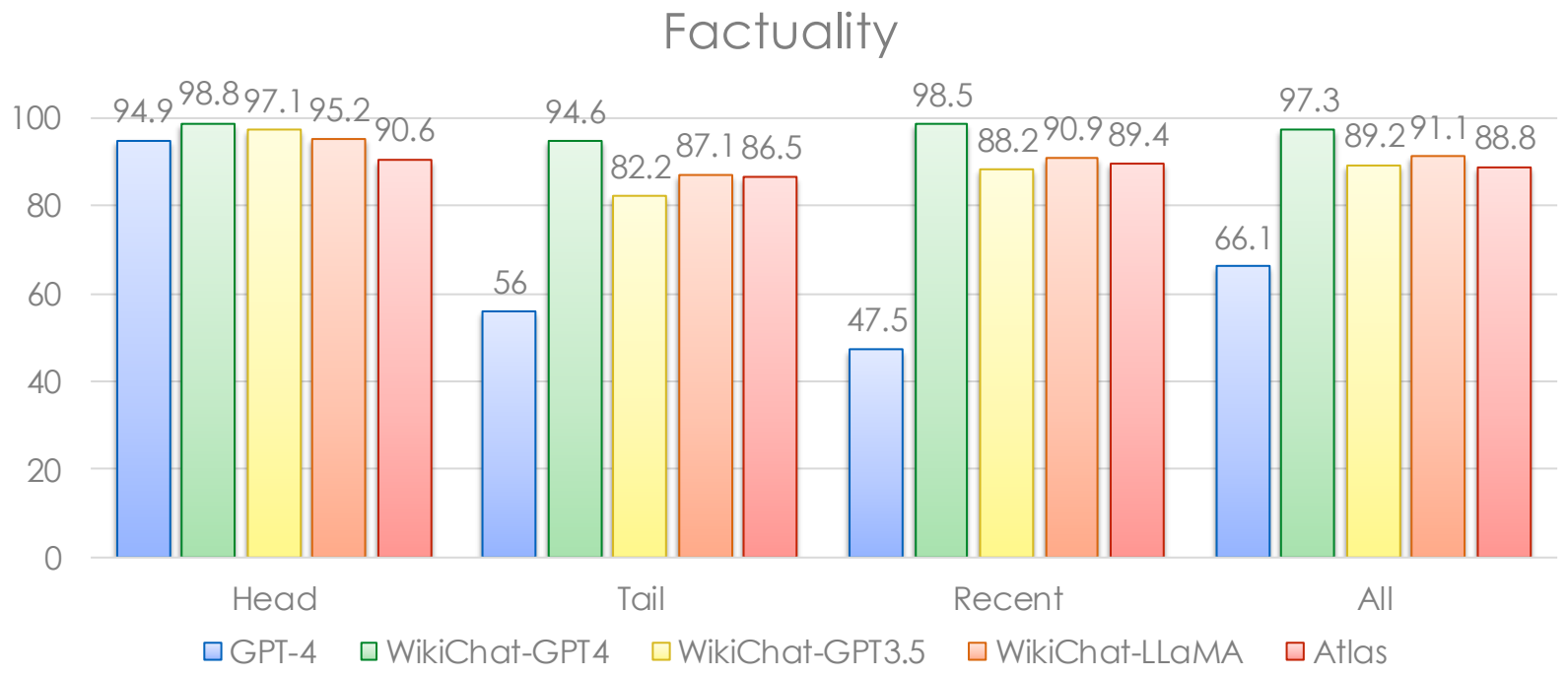
Semnani, Sina et al. WikiChat: Stopping the Hallucination of Large Language Model Chatbots by Few-Shot Grounding on Wikipedia, EMNLP Findings 2023

Simulated Dialogues: WikiChat-LLaMA

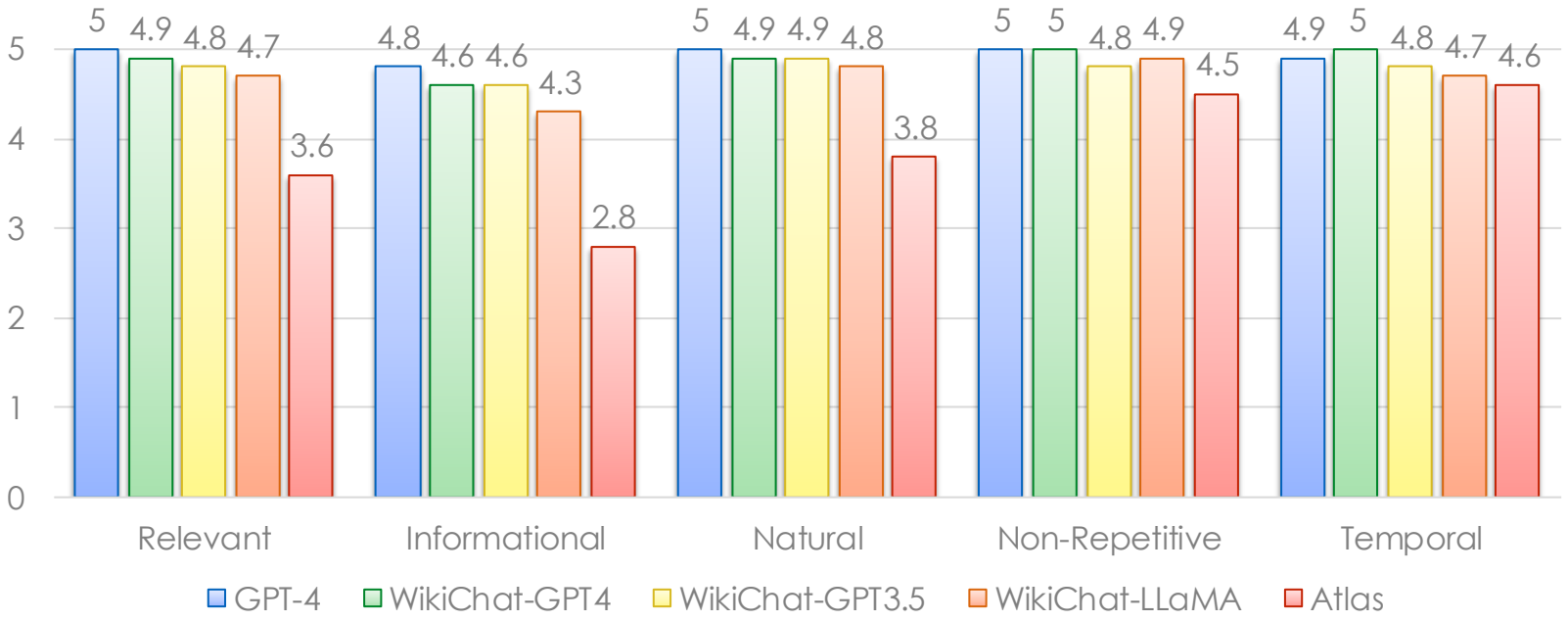


Semnani, Sina et al. WikiChat: Stopping the Hallucination of Large Language Model Chatbots by Few-Shot Grounding on Wikipedia, EMNLP Findings 2023

Simulated Dialogues: Atlas (Contemporary chatbot)



Conversationality



Atlas: Sacrifices conversationality for factuality

WikiChat: maintains conversational with factuality

Design Analysis: Do we need both IR and LLM?

Information Retrieval and LLM are color-coded

User: Can we talk about Sting, the musician?
WikiChat: Sure! Sting is an English musician and actor, best known as the frontman, songwriter and bassist for the new wave rock band the Police. He has released several solo albums and has won numerous awards, including 17 Grammy Awards and a Golden Globe.

User: […] What do you think makes Sting's music so unique?
WikiChat: Sting's music is often very personal and introspective, and he often uses his music to express his own thoughts and feelings. He also has a great ability to blend different genres and create unique sounds.

User: Do you think his birthplace has influenced his music?

WikiChat: Yes, Sting has been heavily influenced by his birthplace of Newcastle upon Tyne, England. As a child, he was inspired by the Queen waving at him from a Rolls-Royce, which diverted him from the shipyard prospect towards a more glamorous life. He was also influenced by the music he heard in local nightclubs he visited such as Club A'Gogo.

Analyzing the Design

- Analysis of each component
  - Do we need 2 pipelines? YES
    - 27% of facts come from LLM generation
  - Do we need to fact-check LLM output: YES
    - Failed: 20% (head); 45% (tail); 56% (recent)

Quiz:

Are We Done Analyzing WikiChat?

Real User Evaluation

- Real user evaluation was seldom performed before LLM because of poor accuracy
- Most important metric!

User study: User reads the first sentence of a new Wikipedia page

Model	User Rating (out of 5)	Factuality
GPT-4	3.4	42.9%
WikiChat using GPT-4	3.8	97.9%

GPT-4: Users are not even aware that over half of the statements are false

Extra Notes

- **Speed**
  - We showed distilling to Llama could speed up the process
  - OpenAI O1-mini is fast enough without fine-tuning
- **Multi-lingual Knowledge**
  - Requires multi-lingual entity resolution
    e.g. Hsiang-Tsung Kung and 孔祥重 is the same person (Wikidata)

How to Build a Persuasion Bot?

donation solicitation

Recommendations

health intervention

Are LLMs Good at Persuasion?

Very persuasive, much better than previous SOTA bots with "persuasion acts"

BUT it hallucinates success stories! "There is a girl called Maria ...."

Is that acceptable?

- **Is Wikichat pipeline sufficient?**
  - **No!** all the success stories are removed! Not very persuasive!
  - Solution: Search with the intent of the fake story:
    - "Tell an impact story about an individual"
- **Is the evaluation method for conversationality applicable?**
  - **No!** LLM evaluation on persuasiveness is not correlated with human evaluation
  - Solution: Manual evaluation

Zero-shot Persuasive Chatbots with LLM-Generated Strategies and Information Retrieval
Kazuaki Furumai, Roberto Legaspi, Julio Vizcarra, Yudai Yamazaki, Yasutaka Nishimura, Sina J. Semnani, Kazushi Ikeda, Weiyan Shi, Monica S. Lam
In Findings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP) , Miami, Florida, November 12–16, 2024.

Conclusion

Stopping hallucination is difficult!

- But it can be done
  - Formalize the human cognitive process step-by-step
    - Systematically create well-tested step-by-step algorithms
    - Optimize for speed later
  - Detailed LLM engineering
    - Development + evaluation (LLM + Real users)
    - Error analysis
- Is it worth it?
  - A critical building block of trustworthy agents!

Agent Engineering Methodology

Systematically create well-tested step-by-step algorithms

1. Design the metrics

2. Start with the most general instruction

3. Debug until you can't break it

- Test with "critical" examples
- Root-cause analysis of LLM errors
- Add/refine the steps
- Add rationale to the prompt; add critical examples as an aid

4. Evaluate

- Tests must cover the hardest cases
- Test with LLM-simulated chats with hard, varied personalities
- Evaluate with an LLM
  - If LLM disagrees with human
    - Use human evaluation
- Test with real users

Good news: Debugging with critical examples achieves high Wikichat quality.

The key: Look at the errors!